UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Geometrical Interpretation of Partial Derivatives

Complete study pack: higher-order partials, partial derivatives as slopes, gradient, concavity and mixed partials, and problems

Compiled from the PESU class 002 slide deck, the class 002 AV-summary transcript (with three targeted video frames), the Unit 1 class-work sheet and past ESA papers. Generated 14 October 2026.

· How to use this pack

Topics 1–7 follow the order of the lecture. Every topic has the same parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S002·7	Deck 002, slide 7 (rendered .pptx)
T002 00:06:10	AV-summary transcript 002, segment start time
V002 00:07:45	Frame grabbed from the class 002 video
CW U1 P2 · ESA Dec-23 1a	Unit 1 class-work problem 2 · past-paper question

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1 Higher-order partial derivatives

Differentiating a partial derivative again gives higher-order partial derivatives. A function of two variables has 2 first-order, 4 second-order and 2^n nth-order partial derivatives; with m variables there are m^n of order n. The two mixed second-order partials are usually equal.

Key terms and definitions

First-order partial derivative. The partial derivative $\partial f/\partial x$ obtained by differentiating once (here, with respect to x while y is held constant). S002·3

Second-order partial derivatives. $\dfrac{\partial^2 f}{\partial x^2}$, $\dfrac{\partial^2 f}{\partial y^2}$, $\dfrac{\partial^2 f}{\partial x \partial y}$, $\dfrac{\partial^2 f}{\partial y \partial x}$, which are obtained by differentiating twice. S002·3

Higher-order partial derivatives. Third, fourth and nth order derivatives involve 3, 4 or n times differentiation respectively. S002·3

Mixed partial derivative. A higher-order partial in which you differentiate with respect to more than one variable, e.g. f_{xy} and f_{yx}. S002·9 · T002 00:00:06

> *"A function of 2 variables has two first order derivatives, four second order derivatives and 2^n of nth order derivatives. A function of m independent variables will have m^n derivatives of order n."* S002·3

Formulas

E1.1 · First-order partial derivatives ESA ★★

$$f_x = \frac{\partial f}{\partial x}, \qquad f_y = \frac{\partial f}{\partial y}$$

gives	Rate of change of f in the x (or y) direction.
applies	Differentiate once; hold the other variable constant.
note	Two of them for a function of two variables.

S002·3 · T002 00:00:44

E1.2 · Second-order partial derivatives ESA ★★

$$f_{xx} = \frac{\partial^2 f}{\partial x^2}, \quad f_{yy} = \frac{\partial^2 f}{\partial y^2}, \quad f_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial y\,\partial x}, \quad f_{yx} = \frac{\partial^2 f}{\partial x\,\partial y}$$

gives	The four second-order partials of $f(x, y)$.
applies	Differentiate twice. In f_{xy}, the subscripts are read left to right: first x, then y.
note	Notation order varies between books. The two mixed partials are equal when continuous (E1.4).

added for clarity S002·3 · T002 00:00:58

E1.3 · How many partial derivatives of order n? ESA ★

$$2 \text{ variables: } 2^n \qquad m \text{ variables: } m^n$$

GIVES The number of nth-order partial derivatives, counting each order of differentiation separately.

APPLIES Two variables: 2 first-order, 4 second-order, 8 third-order.

NOTE With equal mixed partials, two variables give only $n+1$ distinct ones.

S002·3 · T002 00:01:37 · T002 00:02:13

E1.4 · Equality of mixed partials (Clairaut) ESA ★★

$$f_{xy} = f_{yx} \quad \Longleftrightarrow \quad \frac{\partial^2 f}{\partial y\, \partial x} = \frac{\partial^2 f}{\partial x\, \partial y}$$

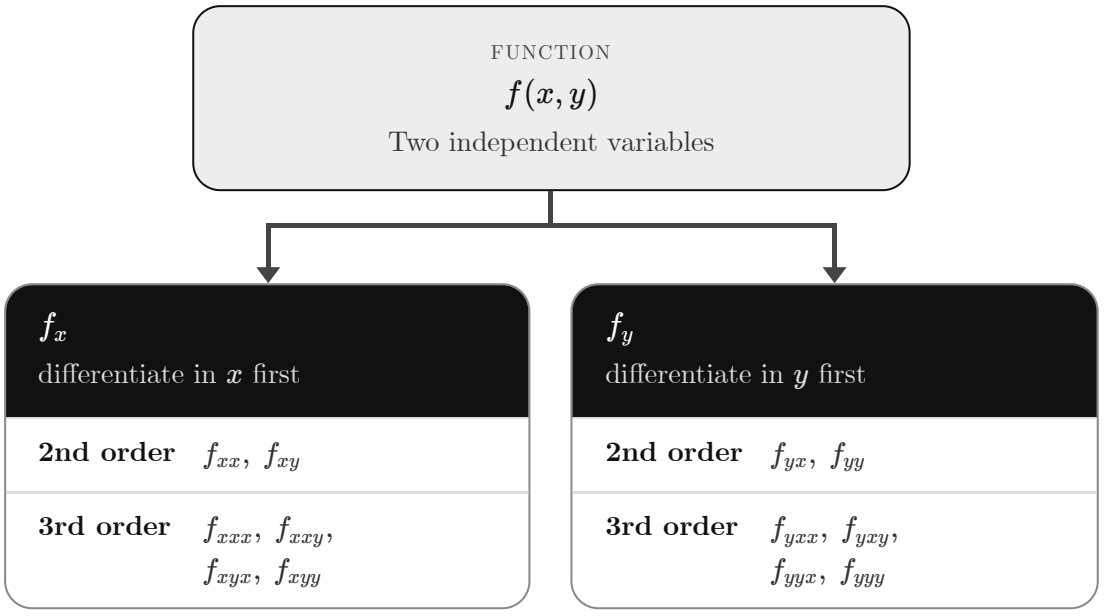
GIVES Order of differentiation does not matter.

APPLIES When f_{xy} and f_{yx} are continuous (true for every function in this unit's problems).

NOTE Verified in CW U1 P2 and P8.

TRANSCRIPT ONLY ADDED FOR CLARITY T002 00:00:24 · T002 00:14:59 · CW U1 P2

Flowcharts



Remark. Totals: 2, 4, 8, ..., 2^n. With m variables there are m^n. When mixed partials are equal, only $n+1$ of the 2^n are different.

Figure F1.1. Counting partial derivatives of each order. ESA ★

S002·3 · T002 00:01:37 · T002 00:02:13

Important points and why

1. A partial derivative is taken with respect to one variable while every other variable is held constant. ESA ★★ T002 00:00:06
2. Differentiating a first-order partial again gives a second-order partial; n differentiations give an nth-order partial. ESA ★★ S002·3
3. For $f(x, y)$: 2 first-order (f_x, f_y), 4 second-order ($f_{xx}, f_{yy}, f_{xy}, f_{yx}$), 2^n of order n. ESA ★★ S002·3 · T002 00:00:44
4. For $f(x_1, \ldots, x_m)$: m^n derivatives of order n; e.g. m^2 second-order partials. ESA ★ S002·3 · T002 00:02:13
5. The two mixed partials "were equal" in the class examples: $f_{xy} = f_{yx}$ for the smooth functions of this unit. ESA ★★ TRANSCRIPT ONLY T002 00:00:24

Exam trap. In f_{xy} the order is x first, then y. Some books write the same derivative as $\partial^2 f / \partial y \, \partial x$. Both mean the same, and for this unit's functions the order does not change the answer. ADDED FOR CLARITY

SHORT-NOTE RECAP

Differentiating $f(x, y)$ once gives the first-order partials f_x, f_y; differentiating again gives four second-order partials f_{xx}, f_{yy}, f_{xy}, f_{yx}. In general n differentiations give 2^n partials of order n for two variables and m^n for m variables. In f_{xy}, differentiate first in x, then in y. When the mixed partials are continuous, $f_{xy} = f_{yx}$ (Clairaut).

2 The one-variable picture: derivative as slope

For $y = f(x)$ the derivative measures change: how fast y changes as you move along the x-axis. Geometrically, $f'(x_0)$ is the slope of the tangent line to the curve at x_0. Partial derivatives extend exactly this idea to surfaces.

Key terms and definitions

Derivative (one variable). For $y = f(x)$, dy/dx is the change in y produced by a change in position along the x-axis; at a point it equals the slope of the tangent to the curve. S002·4 · T002 00:02:50

> *"We know that the derivative of a single valued function gives the slope of the tangent line to the curve at the given point."* S002·4

Formulas

E2.1 · Derivative as slope (one variable) ESA ★

$$\left.\frac{dy}{dx}\right|_{x_0} = f'(x_0) = \text{slope of the tangent to } y = f(x) \text{ at } (x_0, f(x_0))$$

GIVES Slope of the tangent line.

APPLIES f differentiable at x_0.

S002·4 · T002 00:03:13

E2.2 · Tangent line to y = f(x) ESA ★

$$y - f(x_0) = f'(x_0)\,(x - x_0)$$

GIVES Equation of the tangent at x_0.

APPLIES f differentiable at x_0.

NOTE Point-slope form.

ADDED FOR CLARITY S002·4

Graphs and diagrams

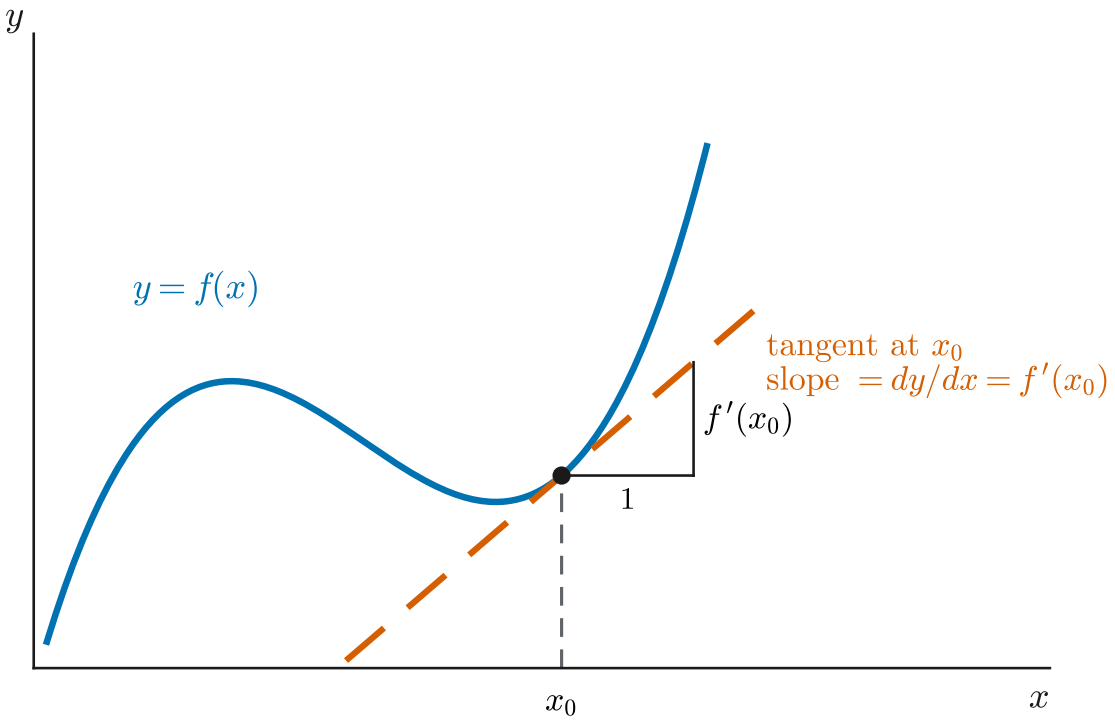


Figure G2.1. Derivative of y = f(x) as the slope of the tangent. ESA ★

S002·4 · T002 00:03:13

How to read it. The blue curve is $y = f(x)$, and the dashed vermilion line is its tangent at x_0. The small triangle shows the slope: a run of 1 gives a rise of $f'(x_0)$. In an exam, label the point, the tangent and the slope dy/dx at x_0.

Important points and why

1. A derivative always represents change: dy/dx is the change on the curve as you change your position on the x-axis. ESA ★ T002 00:02:50 · T002 00:02:59
2. At a particular point, the derivative is the slope of the tangent at that point. ESA ★ S002·4 · T002 00:03:13

SHORT-NOTE RECAP

For $y = f(x)$, dy/dx measures how y changes as x changes. At x_0 it equals the slope of the tangent line to the curve at $(x_0, f(x_0))$, and the tangent is $y - f(x_0) = f'(x_0)(x - x_0)$.

3 A function of two variables is a surface

$z = f(x, y)$ gives one height z for every point (x, y), so its graph is a surface in three dimensions. If (x, y) is restricted to a curve in the xy-plane, such as a circle or a parabola, the matching points of the surface form a curve lying on the surface.

Key terms and definitions

Bivariate function. A function $z = f(x, y)$ of two independent variables. Its graph is a surface in three dimensions. T002 00:04:16

Curve on a surface. If the points (x, y) are restricted to a curve in the xy-plane (circle, parabola, ...), the points $(x, y, f(x, y))$ form a curve that lies on the surface. S002·5

> *"If the points (x,y) lie on a circle in the xy plane, then z = f(x,y) is a curve which looks like the graph shown. If the points (x,y) lie on a parabola in the xy plane, then z = f(x,y) is a curve which looks like the graph shown."* S002·5

Graphs and diagrams

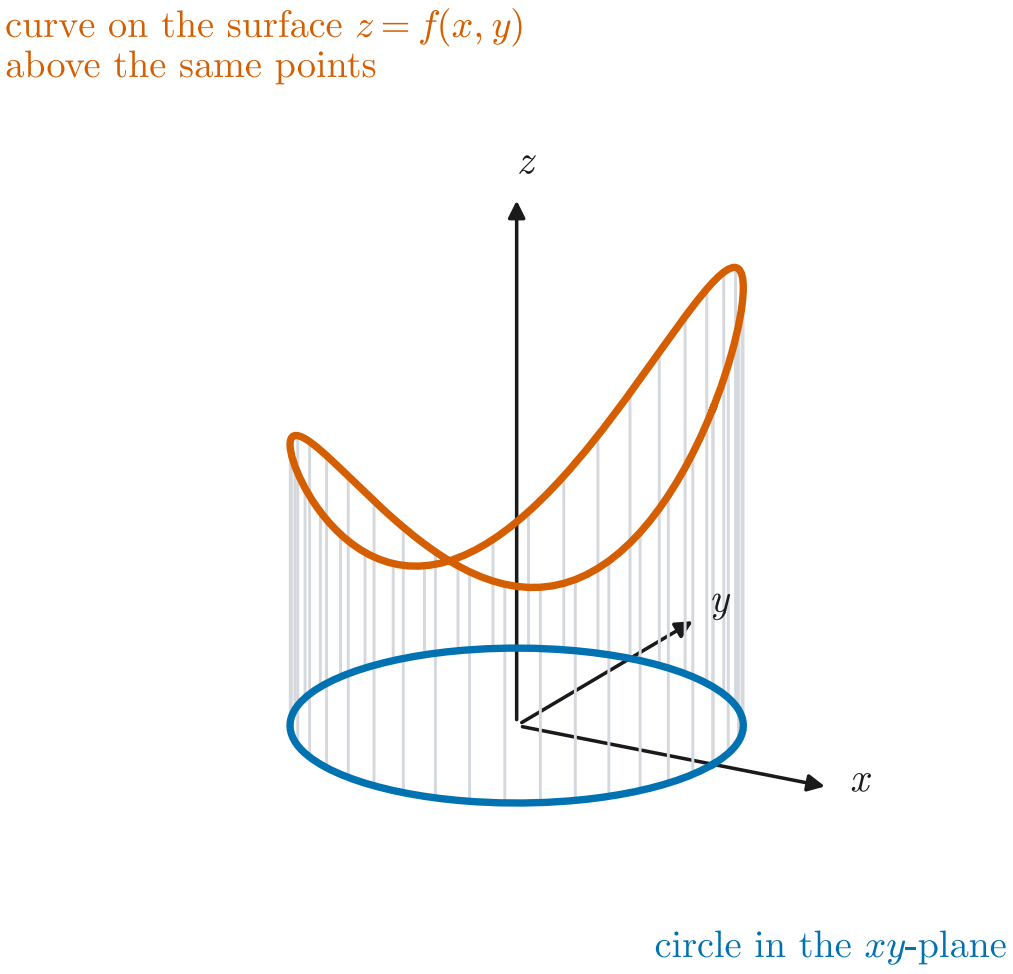


Figure G3.1. Points on a circle in the xy-plane give a curve on the surface. ESA ★

S002·5 · T002 00:03:32

How to read it. The blue circle lies in the xy-plane. Each grey line rises to $z = f(x, y)$, and the vermilion curve joins the tops: a closed curve lying on the surface $z = f(x, y) = 2.2 + 0.8xy + 0.5x$ (an illustrative choice).

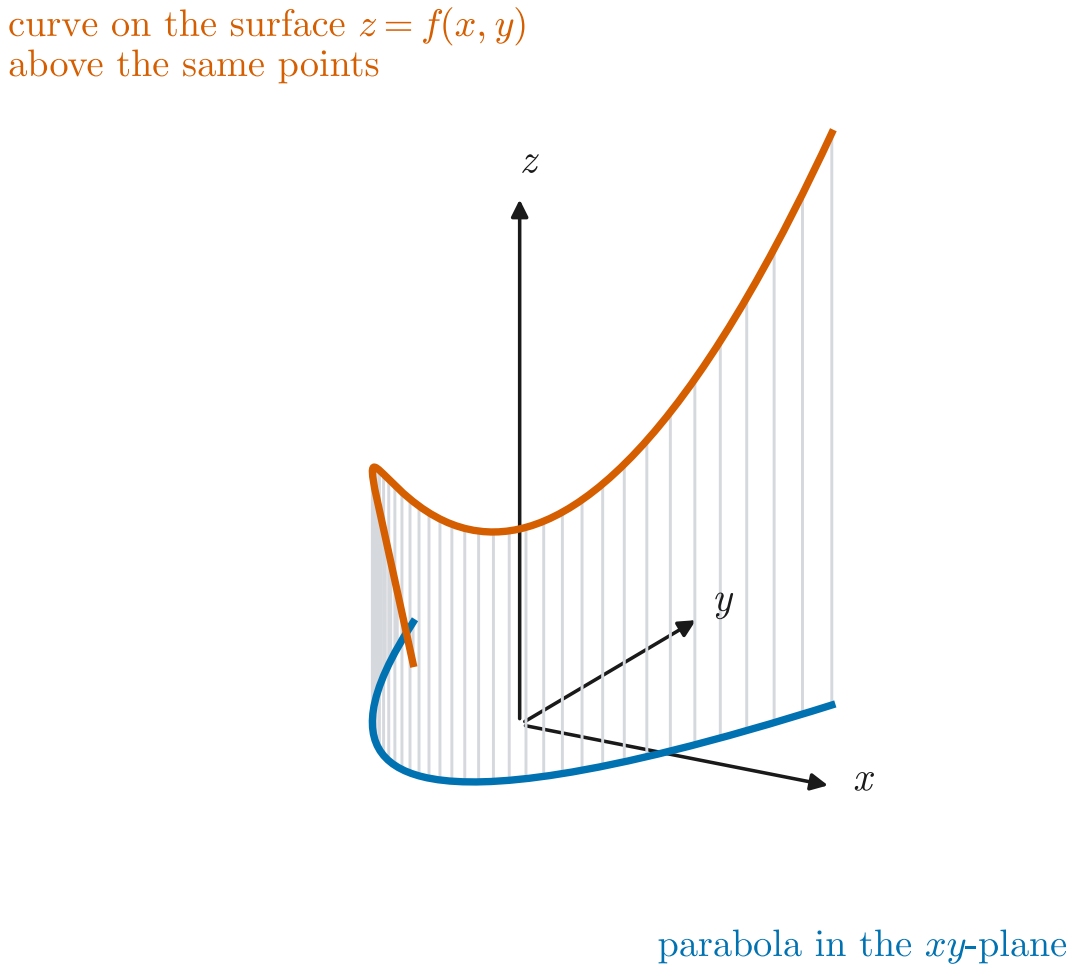


Figure G3.2. Points on a parabola in the xy-plane give a curve on the surface. ESA ★

S002·5 · T002 00:03:47

How to read it. Same idea with a parabola $y = x^2 - 1.2$ in the xy-plane. Every point of the parabola has one height on the surface, so the vermilion curve lies on the surface above it.

Important points and why

1. A function of two independent variables always gives a surface in three dimensions: each (x, y) has one z. ESA ★ T002 00:04:16
2. Restricting (x, y) to a curve in the xy-plane (circle, parabola) picks out a curve on the surface. ESA ★ S002·5 · T002 00:03:32

SHORT-NOTE RECAP

A bivariate function $z = f(x, y)$ assigns one height to each point of the xy-plane, so its graph is a surface in 3-D. Points (x, y) on a circle or parabola in the xy-plane give a curve lying on that surface.

4 First-order partials as slopes of tangents

At a point P on a surface there are infinitely many tangent lines, one for each direction. $\partial f/\partial x$ at (x_0, y_0) is the slope of the one parallel to the x-axis, found by fixing $y = y_0$; $\partial f/\partial y$ is the slope of the one parallel to the y-axis, found by fixing $x = x_0$.

Key terms and definitions

Partial derivative, geometrically. The slope of the tangent line to the surface $z = f(x, y)$ at (x_0, y_0), along the direction of the variable you differentiate with respect to. S002·6

Trace (plane section). The curve cut from the surface by a vertical plane such as $y = y_0$ (giving $z = f(x, y_0)$) or $x = x_0$ (giving $z = f(x_0, y)$). (added for clarity) S002·7

> *"Partial derivative of the function f(x,y) at the point (x_0, y_0) can be visualized as the slope of the tangent line along the direction of the independent variable with respect to which we would differentiate the function."* S002·6

Formulas

E4.1 · ∂f/∂x as an ordinary derivative ESA ★★

$$\left.\frac{\partial f}{\partial x}\right|_{(x_0,y_0)} = \left.\frac{d}{dx}\big(f(x, y_0)\big)\right|_{x=x_0}$$

gives	Slope of the tangent to the surface at $(x_0, y_0, f(x_0, y_0))$ that is parallel to the x-axis.
applies	Fix $y = y_0$ first; $f(x, y_0)$ depends only on x.
note	The tangent lies in the plane $y = y_0$.

S002·7 · V002 00:07:45 · T002 00:06:10

E4.2 · ∂f/∂y as an ordinary derivative ESA ★★

$$\left.\frac{\partial f}{\partial y}\right|_{(x_0,y_0)} = \left.\frac{d}{dy}\big(f(x_0, y)\big)\right|_{y=y_0}$$

gives	Slope of the tangent to the surface that is parallel to the y-axis.
applies	Fix $x = x_0$ first; $f(x_0, y)$ "just depends on y".
note	The tangent lies in the plane $x = x_0$.

S002·7 · V002 00:07:45 · T002 00:07:10

E4.3 · The two tangent lines at P

ESA ★

$$x\text{-direction: } z - z_0 = f_x(x_0, y_0)\,(x - x_0),\quad y = y_0$$
$$y\text{-direction: } z - z_0 = f_y(x_0, y_0)\,(y - y_0),\quad x = x_0$$

GIVES Equations of the tangents whose slopes are f_x and f_y.

APPLIES $z_0 = f(x_0, y_0)$.

NOTE Point-slope form inside each plane section.

ADDED FOR CLARITY S002·7

Graphs and diagrams

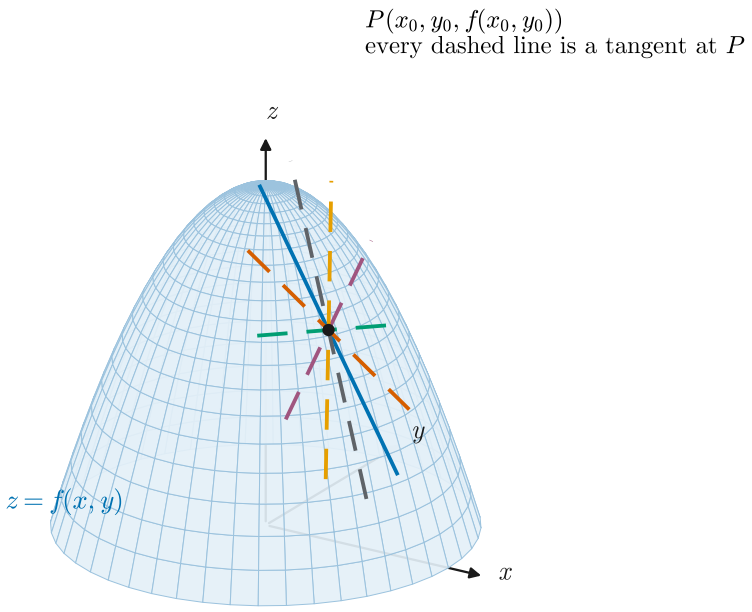


Figure G4.1. A surface has infinitely many tangents at a point. ESA ★★

S002·6 · T002 00:04:36 · T002 00:05:24

How to read it. The dome is $z = 4 - x^2 - y^2$ and $P = (1, -0.5, 2.75)$. Each coloured line is a tangent to the surface at P in a different direction. Partial differentiation picks out two of them: the ones parallel to the x- and y-axes.

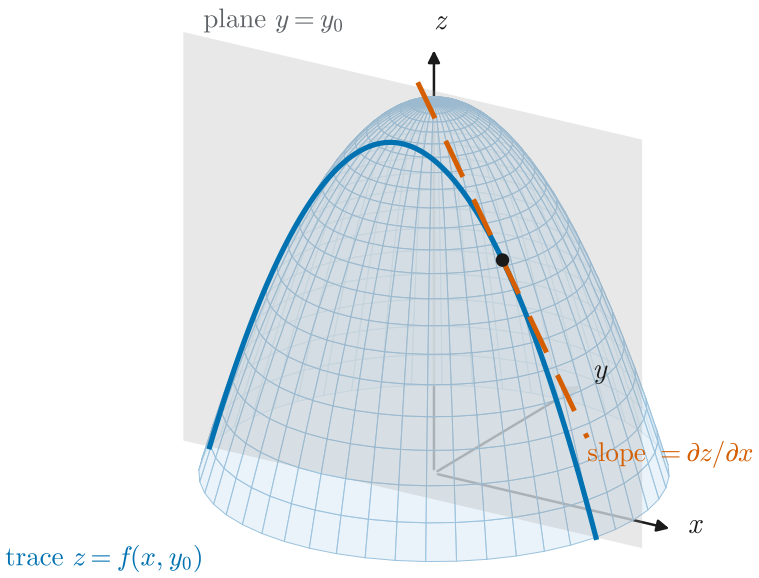


Figure G4.2. $\partial z/\partial x$: the tangent parallel to the x-axis. ESA ★★

S002·7 · V002 00:07:45 · T002 00:06:54

How to read it. The grey plane $y = y_0$ cuts the dome in the blue trace $z = f(x, y_0)$. The dashed tangent to that trace at P is parallel to the x-axis, and its slope is $\partial z/\partial x$ at (x_0, y_0).

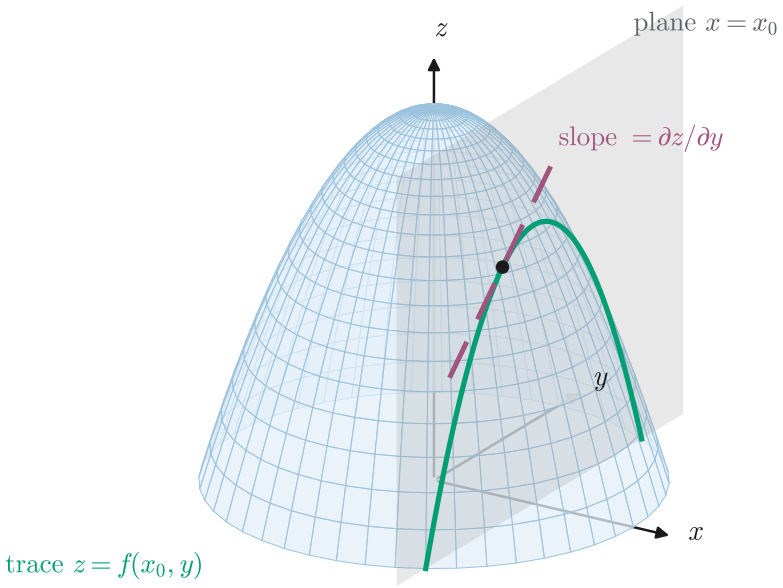


Figure G4.3. $\partial z/\partial y$: the tangent parallel to the y-axis. ESA ★★

S002·7 · V002 00:07:45 · T002 00:07:10

How to read it. The plane $x = x_0$ cuts the dome in the green trace $z = f(x_0, y)$, which "just depends on y". The dashed tangent at P is parallel to the y-axis, and its slope is $\partial z/\partial y$ at (x_0, y_0).

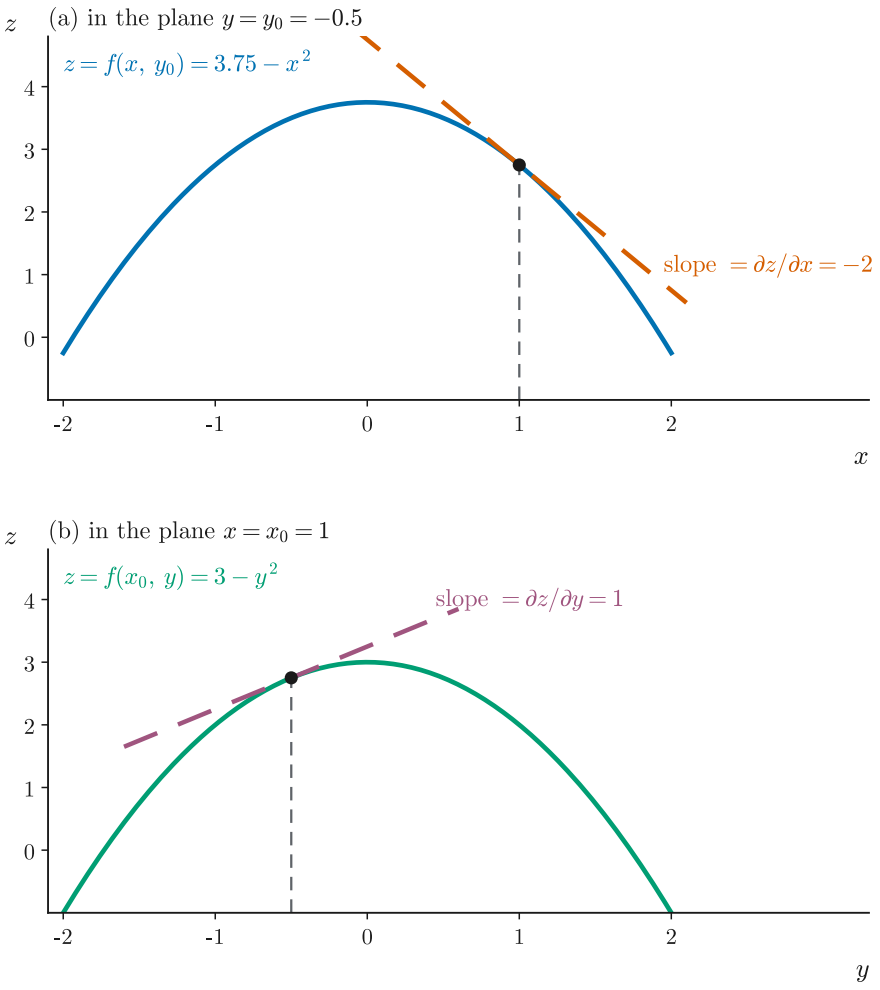
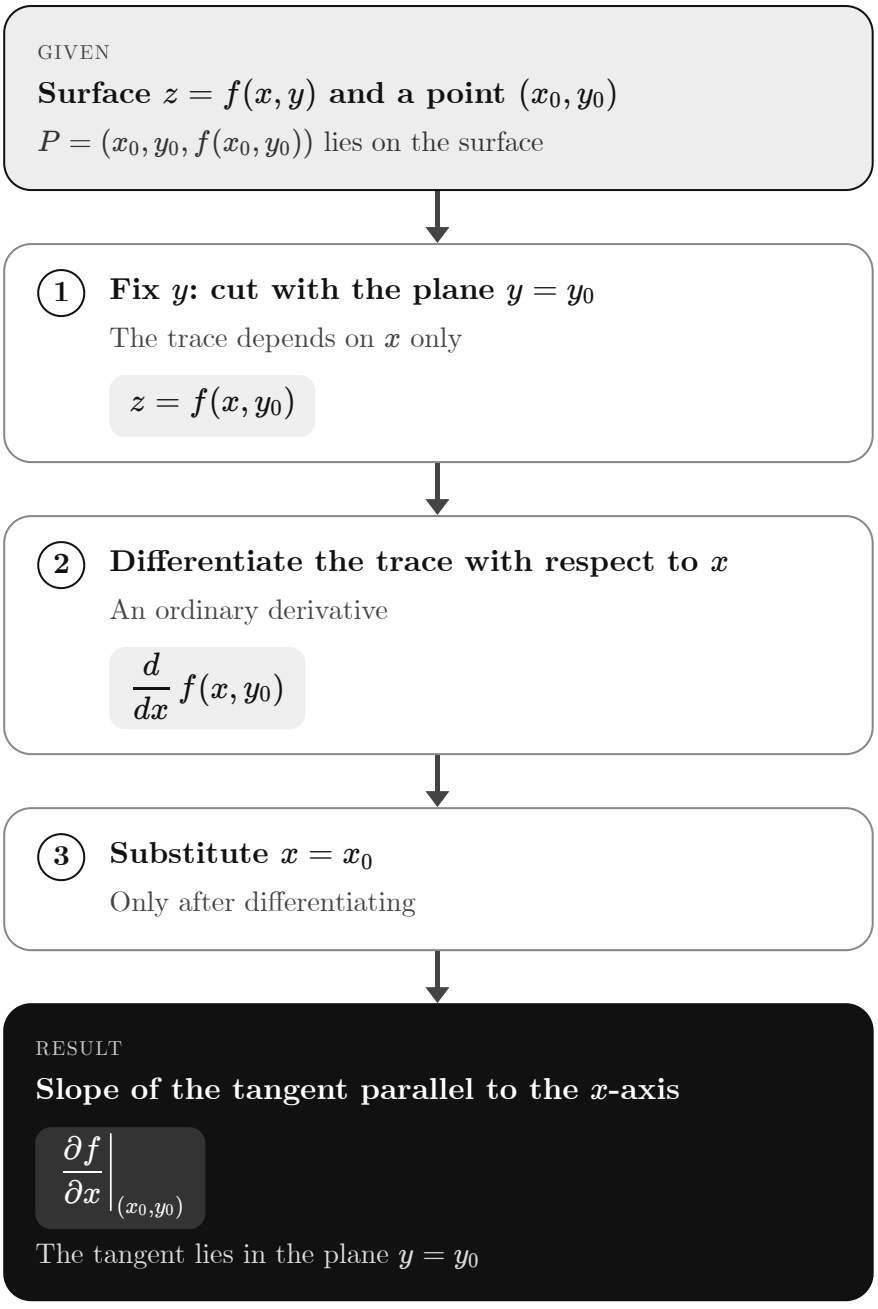


Figure G4.4. The two traces seen flat, with numerical slopes. ESA ★★

ADDED FOR CLARITY S002·7

How to read it. These are panels (a) and (b) from Figures G4.2 and G4.3, drawn in their own planes. At P, the tangent in (a) has slope $-2 = f_x(1, -0.5)$ and the tangent in (b) has slope $1 = f_y(1, -0.5)$. See worked example W4.1.

Flowcharts



Remark. For $\partial f/\partial y$, swap the roles: cut with $x = x_0$, differentiate $f(x_0, y)$ in y, then put $y = y_0$.

Figure F4.1. Partial derivative as the slope of a tangent, step by step. ESA ★★

S002·6 · S002·7 · V002 00:07:45 · T002 00:06:10

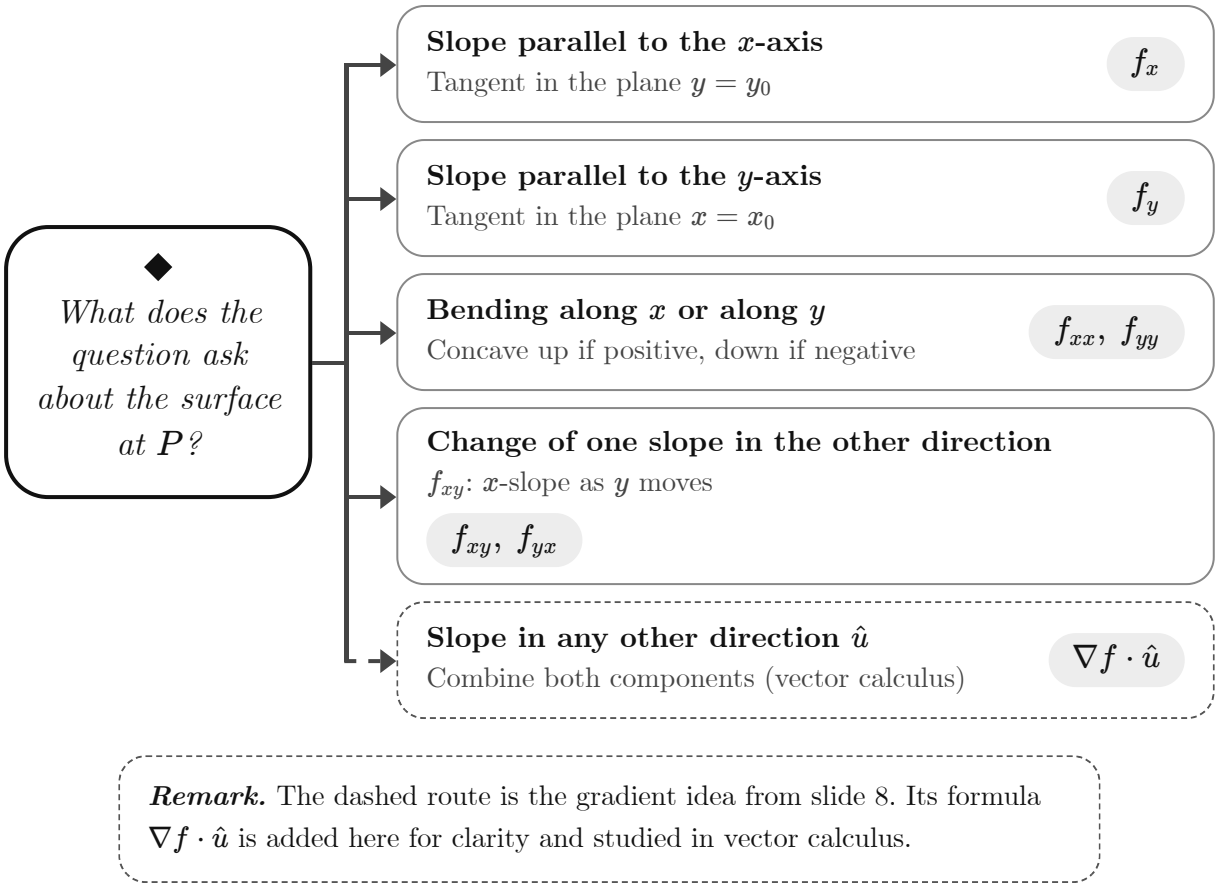


Figure F4.2. Which derivative answers which geometric question. ESA ★★

ADDED FOR CLARITY S002·6 · S002·8 · S002·9 · T002 00:07:54

Important points and why

1. A surface has infinitely many tangents at a given point, one for every direction. ESA ★★ T002 00:05:24 · T002 00:05:35
2. z_x at (x_0, y_0) is the slope of the tangent to the surface at that point which is parallel to the x-axis. ESA ★★ T002 00:06:10 · T002 00:06:27
3. z_y at (x_0, y_0) is the slope of the tangent parallel to the y-axis. ESA ★★ T002 00:07:10 · T002 00:07:54
4. To find $\partial f/\partial x$ at (x_0, y_0): put $y = y_0$, which gives a curve $f(x, y_0)$ in one variable, then differentiate it at $x = x_0$. ESA ★★ S002·7
5. The cutting plane $y = y_0$ slices the surface in the curve whose slope is f_x; the plane $x = x_0$ gives the curve whose slope is f_y. ESA ★★ S002·7 · V002 00:07:45

Why are there infinitely many tangents at one point of a surface?
From the point you can move in infinitely many directions in the xy-plane. Each direction traces a different curve on the surface, and each curve has its own tangent at P.
T002 00:04:36 · T002 00:05:24

Why does fixing $y = y_0$ turn $\partial f/\partial x$ into an ordinary derivative?
Once $y = y_0$, $f(x, y_0)$ depends on x alone. It is a plane curve in the plane $y = y_0$, and its ordinary derivative at x_0 is the slope of its tangent.

S002·7

Exam trap. "Parallel to the x-axis" describes the tangent's direction seen from above (it lies in the plane $y = y_0$). It does not mean the tangent is horizontal: its slope is $f_x(x_0, y_0)$, which is usually not zero. ADDED FOR CLARITY

Exam trap. Hold the other variable constant before differentiating. Do not substitute $x = x_0$ before differentiating with respect to x. ADDED FOR CLARITY

Worked examples

W4.1 · Slopes of the two tangents on a dome

ESA ★★

GIVEN	$f(x, y) = 4 - x^2 - y^2$ (the dome in Figures G4.1–G4.4), point $(x_0, y_0) = (1, -0.5)$.
STEP 1	Height: $z_0 = 4 - 1 - 0.25 = 2.75$, so $P = (1, -0.5, 2.75)$.
STEP 2	Fix $y = -0.5$: $f(x, -0.5) = 3.75 - x^2$, so $\dfrac{d}{dx}(3.75 - x^2) = -2x = -2$ at $x = 1$.
STEP 3	Fix $x = 1$: $f(1, y) = 3 - y^2$, so $\dfrac{d}{dy}(3 - y^2) = -2y = 1$ at $y = -0.5$.
STEP 4	Read: moving parallel to the x-axis the surface falls 2 units per unit step; parallel to the y-axis it rises 1.
ANSWER	$\boxed{f_x(1, -0.5) = -2, \quad f_y(1, -0.5) = 1}$
CHECK	Directly: $f_x = -2x$ and $f_y = -2y$ give the same values.

S002·6 · S002·7

SHORT-NOTE RECAP

A surface has infinitely many tangents at a point $P(x_0, y_0, f(x_0, y_0))$. $\partial f/\partial x$ at (x_0, y_0) is the slope of the tangent parallel to the x-axis. It is found by cutting the surface with the plane $y = y_0$ and differentiating the curve $f(x, y_0)$ at x_0. Likewise $\partial f/\partial y$ is the slope of the tangent parallel to the y-axis, from the curve $f(x_0, y)$ in the plane $x = x_0$.

5 Slope in any other direction: the gradient

A tangent that is parallel to neither axis is handled by resolving the direction into x- and y-components. Treating f_x and f_y as vector components gives their resultant, the gradient ∇f. It is studied fully in vector calculus.

Key terms and definitions

Gradient. The vector $\nabla f = f_x\,\mathbf{i} + f_y\,\mathbf{j}$: the resultant of the two slope components along the x- and y-axes. S002·8 · T002 00:09:23

Directional derivative. The slope of the surface in the direction of a unit vector $\hat{u}$: $D_{\hat{u}} f = \nabla f \cdot \hat{u}$. It is studied in vector calculus. (added for clarity) T002 00:09:54

> *"Slope of tangent along any arbitrary direction (Gradient of a function). What about other directions than parallel to x and y?"* S002·8

Formulas

E5.1 · Gradient ESA ★

$$\nabla f = f_x\,\mathbf{i} + f_y\,\mathbf{j}$$

gives	Vector whose components are the slopes along the x- and y-axes.
applies	Taken at the point (x_0, y_0).
note	Covered fully in vector calculus.

S002·8 · T002 00:09:23

E5.2 · Slope along any direction (directional derivative) ESA ★

$$D_{\hat{u}} f = \nabla f \cdot \hat{u} = a\,f_x + b\,f_y, \qquad \hat{u} = (a, b),\ a^2 + b^2 = 1$$

gives	Slope of the tangent in the direction $\hat{u}$.
applies	$\hat{u}$ must be a unit vector.
note	$\hat{u} = \mathbf{i}$ gives f_x; $\hat{u} = \mathbf{j}$ gives f_y. The largest slope is $\lvert\nabla f\rvert$.

(added for clarity) S002·8

Graphs and diagrams

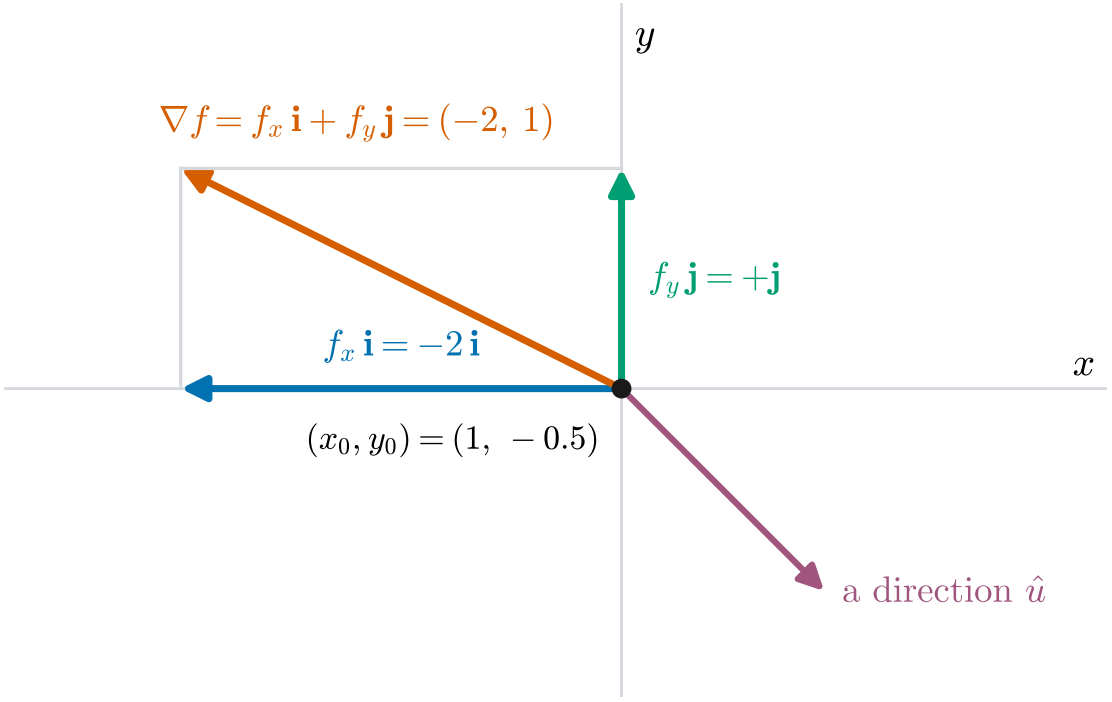
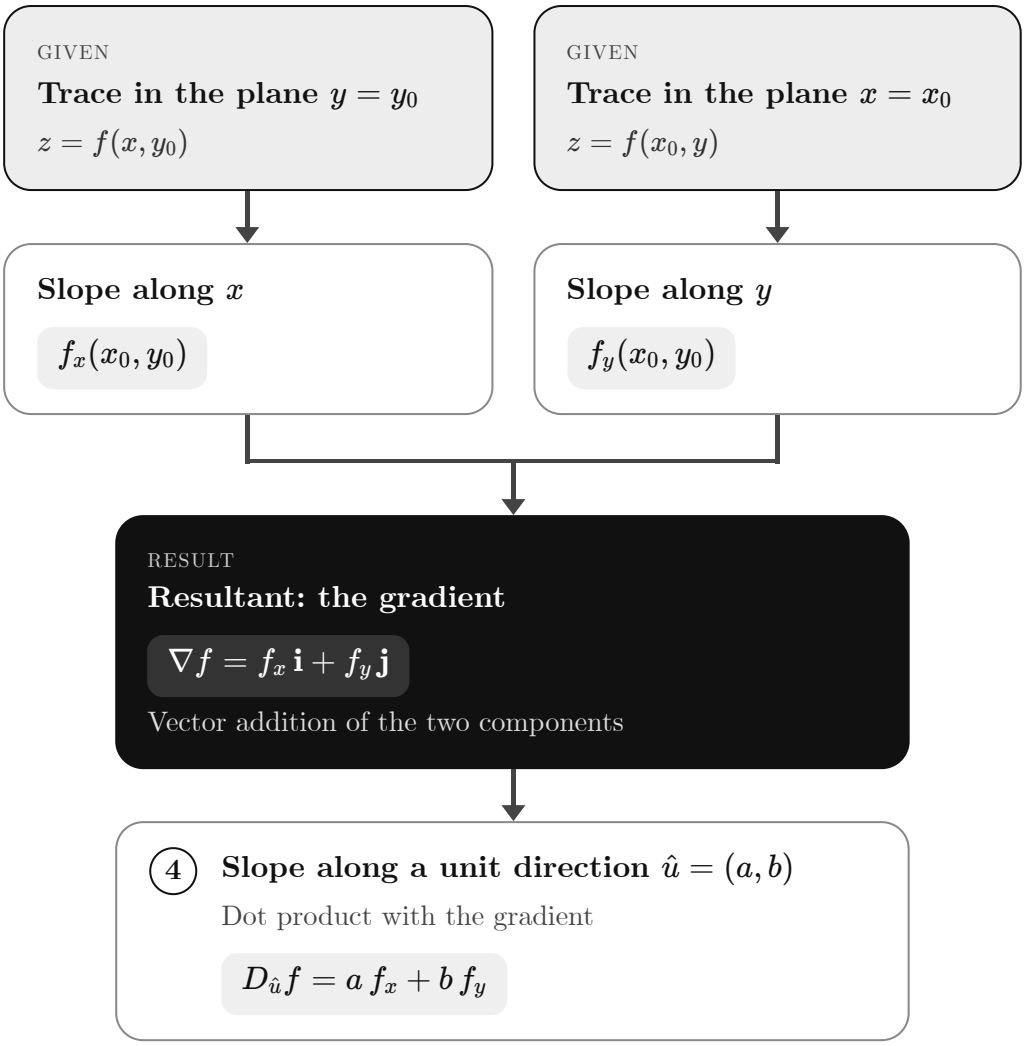


Figure G5.1. Resolving into components: the gradient. ESA ★

S002·8 · T002 00:09:06 · V002 00:09:05

How to read it. The slope components $f_x\,\mathbf{i}$ (blue) and $f_y\,\mathbf{j}$ (green) add by vector addition to give the resultant ∇f (vermilion), drawn here for the dome at $(1, -0.5)$. The purple arrow is another direction $\hat{u}$; the slope along it is $\nabla f \cdot \hat{u}$ (W5.1).

Flowcharts



Remark. Check $\hat{u} = (1, 0)$: you get back f_x. The steepest slope is $|\nabla f| = \sqrt{f_x^2 + f_y^2}$. Example: the dome at $(1, -0.5)$ gives $\nabla f = (-2, 1)$ (W5.1).

Figure F5.1. Two slopes combine into the gradient. ESA ★

ADDED FOR CLARITY S002·8 · T002 00:09:06 · T002 00:09:23

Important points and why

1. f_x and f_y only give slopes in two directions; most tangents are parallel to neither axis. ESA ★ S002·8 · T002 00:08:22
2. Any line in the plane can be resolved into components along the x- and y-axes; vector addition of the components gives the resultant. ESA ★ T002 00:09:06 · T002 00:09:23
3. The resultant of $f_x\,\mathbf{i}$ and $f_y\,\mathbf{j}$ is the gradient ∇f, studied in vector calculus. ESA ★ S002·8 · T002 00:09:44
4. The slope in a unit direction $\hat{u} = (a, b)$ is $af_x + bf_y$. It is greatest, equal to $|\nabla f|$, in the direction of ∇f. ESA ★ ADDED FOR CLARITY S002·8

Why can't f_x or f_y alone give the slope along a diagonal direction?
Each is the slope along one axis only. A diagonal direction has both an x- and a y-component, so both slopes contribute. Combining them gives the gradient, and its dot product with the unit direction gives the slope.

S002·8 · T002 00:09:06

Worked examples

W5.1 · Gradient and the slope along a diagonal ESA ★

GIVEN Same dome and point as W4.1; direction $\hat{u} = \frac{1}{\sqrt{2}}(1, -1)$.

STEP 1 $\nabla f = f_x\,\mathbf{i} + f_y\,\mathbf{j} = -2\,\mathbf{i} + \mathbf{j}$.

STEP 2 $D_{\hat{u}} f = \frac{1}{\sqrt{2}}(-2) + \left(-\frac{1}{\sqrt{2}}\right)(1) = -\frac{3}{\sqrt{2}}$.

STEP 3 Steepest slope: $|\nabla f| = \sqrt{4+1} = \sqrt{5}$, in the direction of ∇f.

ANSWER $\boxed{D_{\hat{u}} f = -3/\sqrt{2} \approx -2.121, \quad |\nabla f| = \sqrt{5} \approx 2.236}$

CHECK $|D_{\hat{u}} f| \le |\nabla f|$, as it must be.

S002·8

SHORT-NOTE RECAP

Tangents in other directions are handled by resolving the direction into x- and y-components. The resultant of $f_x\,\mathbf{i}$ and $f_y\,\mathbf{j}$ is the gradient ∇f. The slope along a unit vector $\hat{u}$ is $\nabla f \cdot \hat{u}$, and the largest slope is $|\nabla f|$.

6 Second-order partials: concavity and mixed partials

As in one variable, the second derivative describes bending. $f_{xx} > 0$ means the surface is concave up in the x direction, and $f_{yy} > 0$ means concave up in the y direction. The mixed partial f_{xy} measures how the x-slope changes as you move in the y direction.

Key terms and definitions

Concave up in the x direction. The trace in the x direction bends upward; the x-slope increases as x increases. Indicated by $f_{xx} > 0$. S002·9

Mixed partial f_{xy}. Tells how the rate of change of $f(x, y)$ in the x direction is changing as we move in the y direction. S002·9

> *"If $f_{xx} > 0$ then f(x,y) is concave up in the x direction. If $f_{yy} > 0$ then f(x,y) is concave up in the y direction. Mixed partials tell us how a partial in one variable is changing in the direction of the other. f_{xy} tells us how the rate of change of f(x,y) in the x direction is changing as we move in the y direction."* S002·9

Formulas

E6.1 · Concavity from second partials ESA ★★

$$f_{xx} > 0 \Rightarrow \text{concave up in } x, \qquad f_{yy} > 0 \Rightarrow \text{concave up in } y$$

gives	Shape of the surface along each axis direction.
applies	$f_{xx} < 0$ (or $f_{yy} < 0$) gives concave down in that direction.
note	Same test as d^2y/dx^2 for one variable.

S002·9 · T002 00:11:19

E6.2 · Mixed partial as a change of slope ESA ★★

$$f_{xy} = \frac{\partial}{\partial y}(f_x)$$

gives	How the x-slope f_x changes as you move in the y direction.
applies	Likewise $f_{yx} = \partial(f_y)/\partial x$: the change of the y-slope along x.

S002·9 · T002 00:11:45

E6.3 · Names of higher derivatives (motion) ESA ★

$$\frac{dy}{dt} = \text{velocity}, \quad \frac{d^2y}{dt^2} = \text{acceleration}, \quad \frac{d^3y}{dt^3} = \text{jerk}$$

GIVES Physical meaning when the independent variable is time.

APPLIES One-variable analogy only.

NOTE Names exist up to about the 6th derivative.

TRANSCRIPT ONLY T002 00:10:23 · T002 00:10:57

Graphs and diagrams

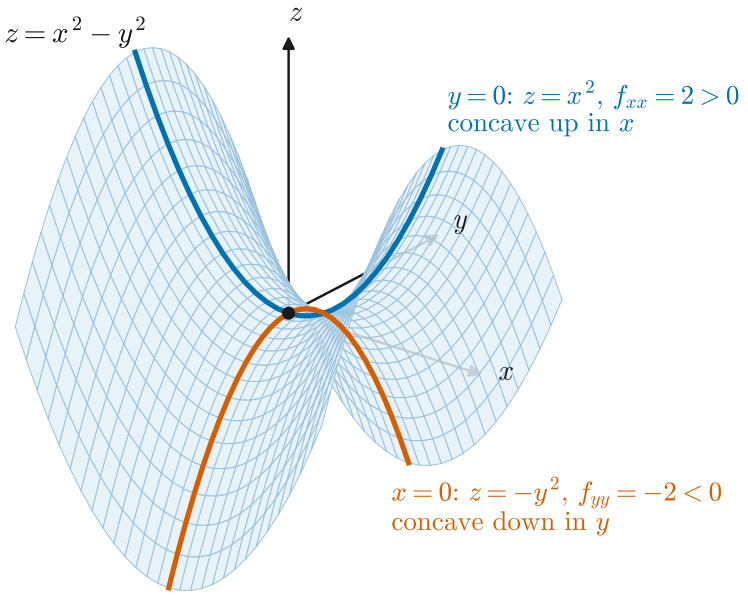


Figure G6.1. Concave up in x, concave down in y: the saddle z = x² − y². ESA ★★

ADDED FOR CLARITY S002·9 · S002·10 · T002 00:11:19

How to read it. Along the x direction (blue trace, $y = 0$) the surface bends up, so $f_{xx} = 2 > 0$. Along the y direction (vermilion trace, $x = 0$) it bends down, so $f_{yy} = -2 < 0$. One point can be concave up in one direction and concave down in another.

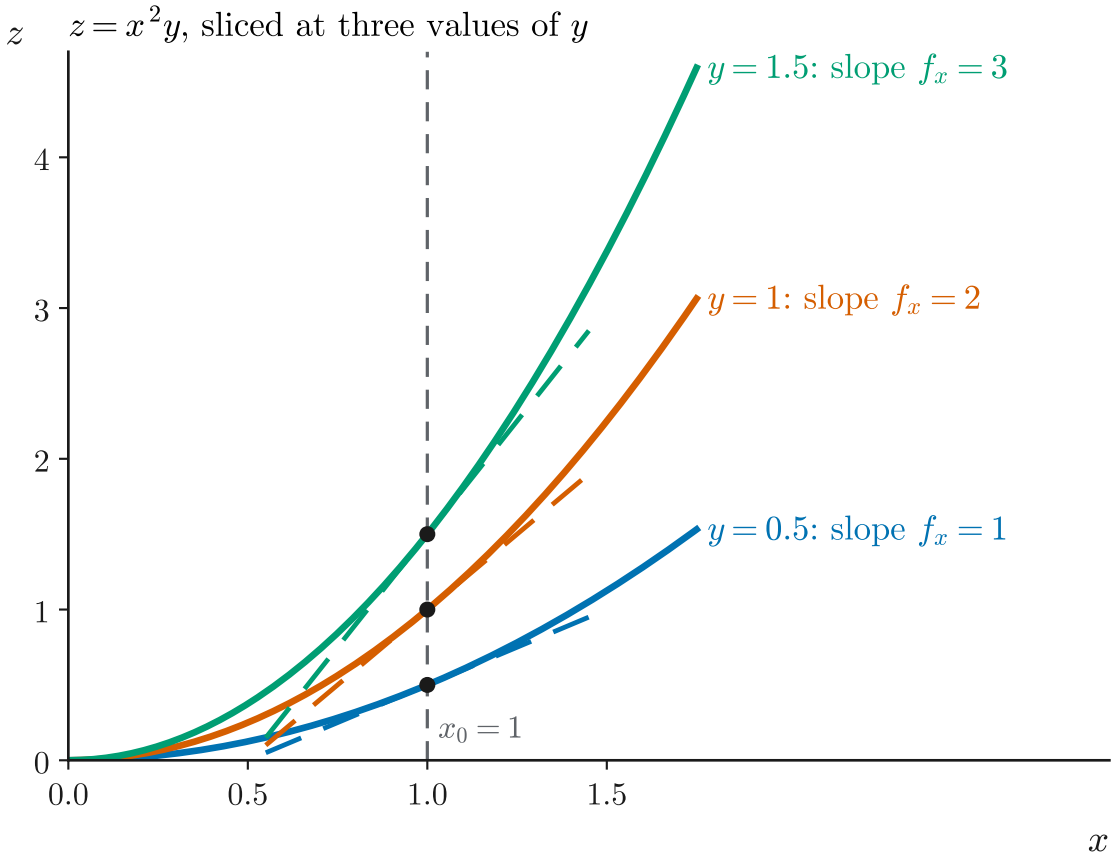
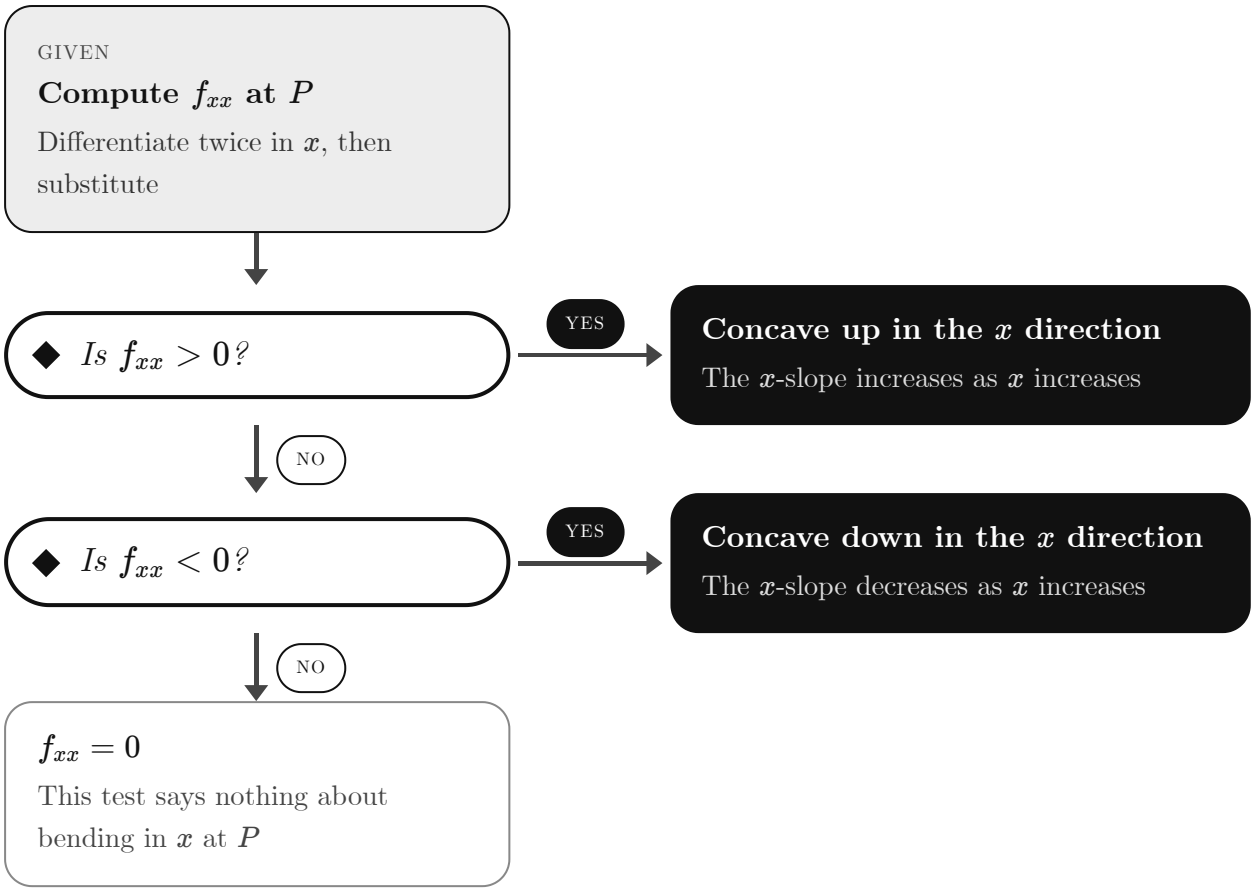


Figure G6.2. Mixed partial: how the x-slope changes as y changes. ESA ★★

ADDED FOR CLARITY S002·9 · T002 00:11:45

How to read it. Three traces of $z = x^2 y$, at $y = 0.5$, 1 and 1.5. At $x_0 = 1$ their tangents have slopes $f_x = 1, 2, 3$. The x-slope grows by 2 per unit of y, which is $f_{xy} = 2$.

Flowcharts



Remark. Run the same ladder with f_{yy} for the y direction. Read the sign of f_{xy} in the same way: positive means the x-slope increases as y increases (Figure G6.2).

Figure F6.1. Reading concavity from a second-order partial. ESA ★★

ADDED FOR CLARITY S002·9 · T002 00:11:19 · T002 00:11:38

One variable vs two variables

	One variable, $y = f(x)$	**Two variables, $z = f(x, y)$**
Graph	Curve in a plane	Surface in 3-D
Tangents at a point	One	Infinitely many, one per direction
First derivative	dy/dx: slope of the tangent	f_x, f_y: slopes parallel to the x- and y-axes; ∇f for other directions
Second derivative	d^2y/dx^2: concave up or down	f_{xx}, f_{yy}: concavity in x, y; f_{xy}, f_{yx}: change of one slope along the other direction
Number of order n	1	2^n (or m^n for m variables)

ADDED FOR CLARITY S002·4 · S002·6 · S002·9 · S002·3

Important points and why

1. As in one variable (concave up or down from d^2y/dx^2), the second-order partials describe how the surface bends. ESA ★★ T002 00:11:03 · T002 00:11:19
2. $f_{xx} > 0$ means concave up in the x direction; $f_{yy} > 0$ means concave up in the y direction. ESA ★★ S002·9 · T002 00:11:38
3. $f_{xx} < 0$ (or $f_{yy} < 0$) means concave down in that direction. One surface can be up in x and down in y, as in a saddle. ESA ★★ ADDED FOR CLARITY S002·9
4. f_{xy} is the change in f_x along the y direction; f_{yx} is the change in f_y along the x direction. ESA ★★ S002·9 · T002 00:11:45 · T002 00:12:04
5. The class demo, "Second-Order Partial Derivatives", lets you set the signs of f_{xx}, f_{yy} and f_{xy} and watch the red tangent line move in 3D and contour views. ESA ★ S002·10 · V002 00:12:40 · T002 00:12:40

Why does $f_{xx} > 0$ mean concave up in x?
f_{xx} is the rate of change of the x-slope as x increases. If it is positive, the tangent keeps tilting upward as you move along x, so the trace bends upward.
S002·9 · T002 00:11:19

Why do the 4 second-order partials usually give only 3 different values?
For the smooth functions used in this unit the mixed partials are equal, $f_{xy} = f_{yx}$ (Clairaut), leaving only f_{xx}, f_{yy} and f_{xy}.
ADDED FOR CLARITY T002 00:00:24

Worked examples

W6.1 · Concave up in x, concave down in y

ESA ★★

given $f(x, y) = x^2 - y^2$ at the origin (Figure G6.1).

step 1 $f_x = 2x$, $f_{xx} = 2 > 0$: concave up in the x direction.

step 2 $f_y = -2y$, $f_{yy} = -2 < 0$: concave down in the y direction.

step 3 $f_{xy} = \partial(2x)/\partial y = 0$: the x-slope does not change as y changes.

answer Saddle: up along x, down along y

check Traces $z = x^2$ (at $y = 0$) and $z = -y^2$ (at $x = 0$) agree.

S002·9

W6.2 · Reading a mixed partial

ESA ★★

given $f(x, y) = x^2 y$ at $x_0 = 1$ (Figure G6.2).

step 1 $f_x = 2xy$, so at $x = 1$ the x-slope is $2y$: 1, 2, 3 for $y = 0.5, 1, 1.5$.

step 2 $f_{xy} = \partial(2xy)/\partial y = 2x = 2$ at $x = 1$.

step 3 Each unit step in y raises the x-slope by 2, which matches the slopes $1 \to 2 \to 3$ for steps of 0.5.

answer $f_{xy}(1, y) = 2$

check $f_{yx} = \partial(x^2)/\partial x = 2x = 2$ as well.

S002·9

short-note recap

Second-order partials describe bending. If $f_{xx} > 0$ the surface is concave up in the x direction; if $f_{yy} > 0$ it is concave up in the y direction (negative means concave down). The mixed partial f_{xy} tells how the rate of change of f in the x direction changes as we move in the y direction.

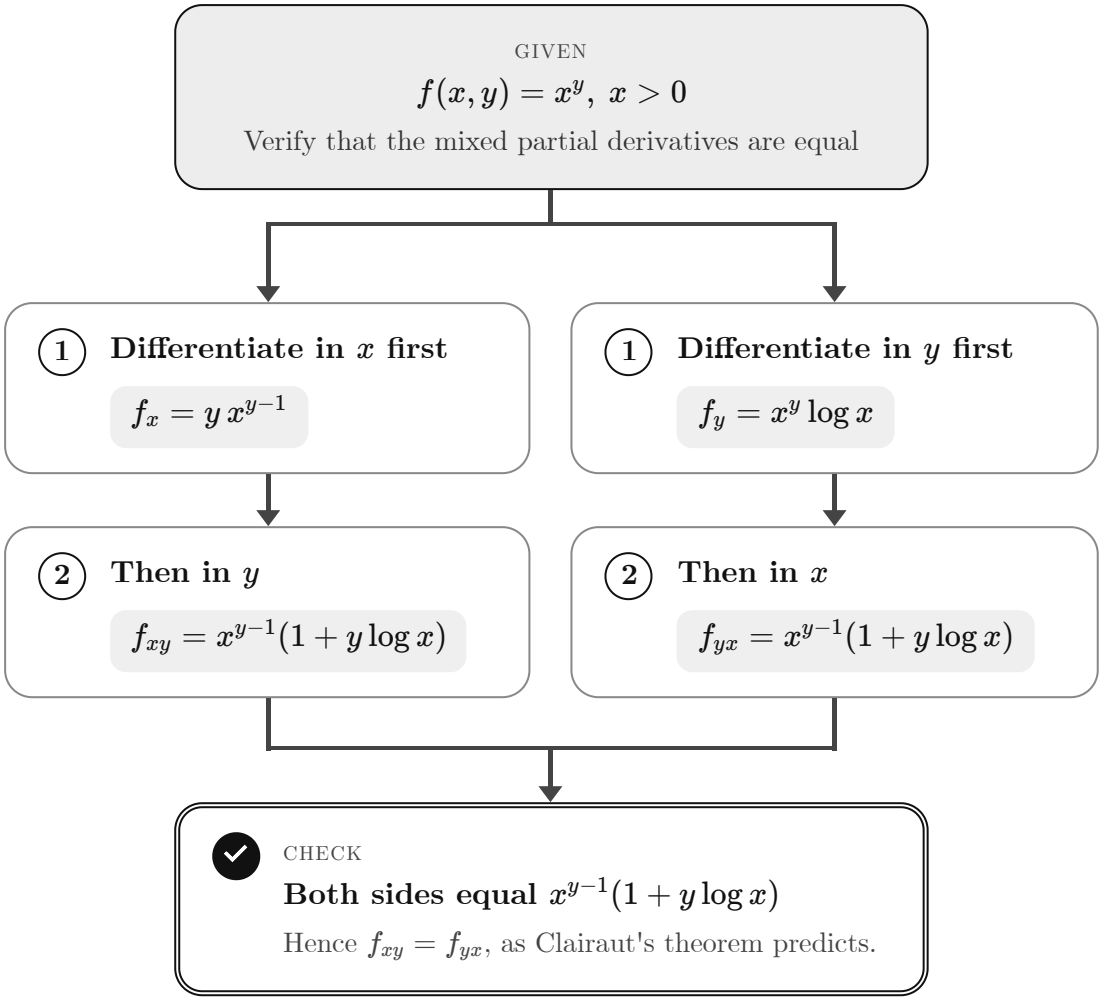
7 Problems on partial derivatives

The deck's title includes “Problems on partial derivatives”, but no problem slides are in it, and the transcript defers the problems to the next class. The problems here come from the Unit 1 class-work sheet and past ESA papers that test exactly this session's content: first-order partials at a point, higher-order partials, and equality of mixed partials.

Key terms and definitions

Clairaut's theorem. If f_{xy} and f_{yx} are continuous near a point, they are equal there. The class plans to verify it in the next session. TRANSCRIPT ONLY ADDED FOR CLARITY T002 00:14:59

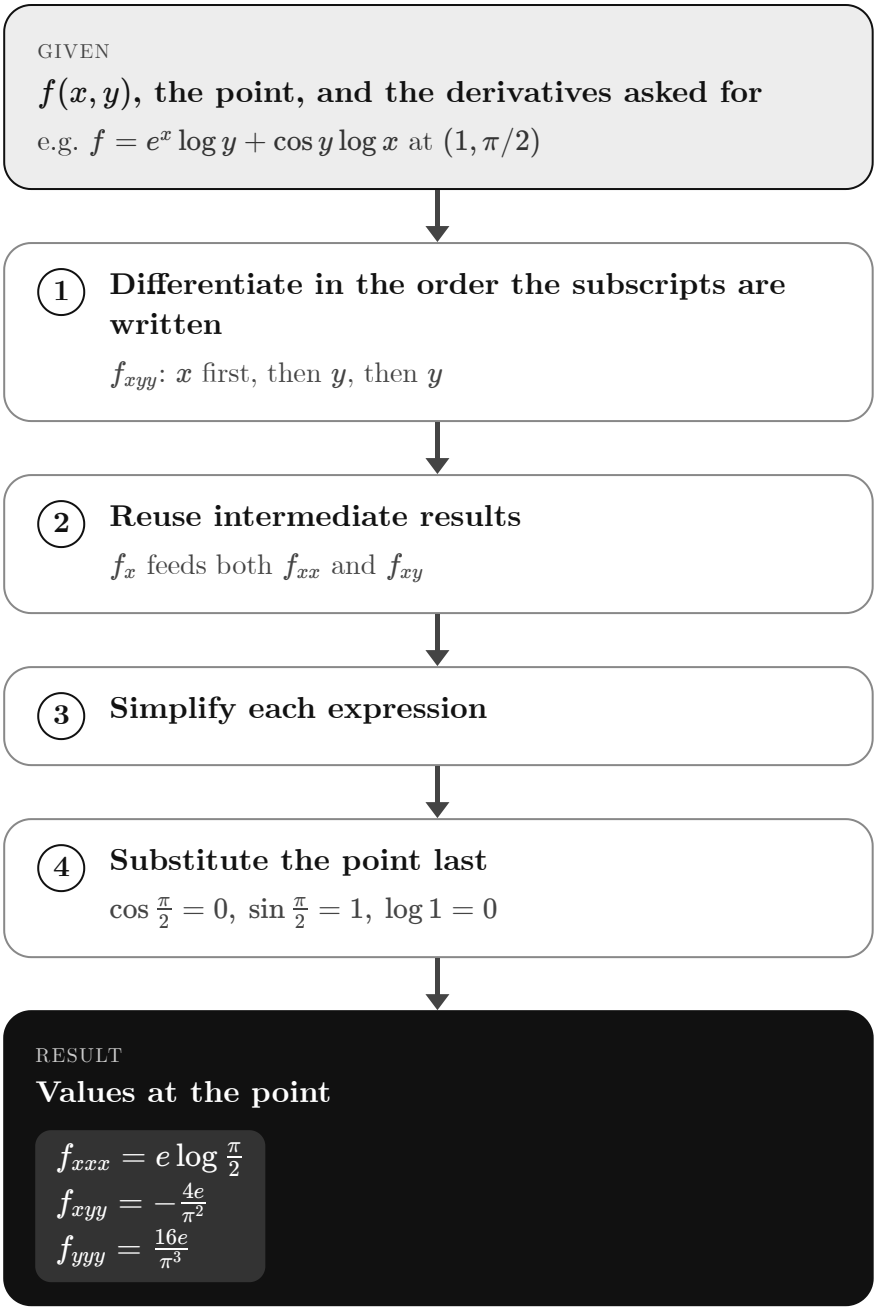
Flowcharts



Exam trap. In x, x^y is a power; in y, it is an exponential. Using the wrong rule is the usual slip.

Figure F7.1. Verifying equal mixed partial derivatives. ESA ★★

CW U1 P2 · T002 00:14:59



Remark. log is the natural logarithm, so $\frac{d}{dy}\log y = \frac{1}{y}$. Check one value against the CW answer key if you can.

Figure F7.2. Evaluating higher-order partials at a point (ESA Dec 2023 Q1.a). ESA ★★

ESA Dec-23 1a · CW U1 P5 · S002·3

Important points and why

1. Evaluate at the point only after all the differentiations are done. ESA ★★ ADDED FOR CLARITY CW U1 P1 · ESA Dec-23 1a
2. To verify $f_{xy} = f_{yx}$, work out each side separately and compare them. ESA ★★ CW U1 P2

Exam trap. In these problems $\log$ is the natural logarithm: $\frac{d}{dy}\log y = \frac{1}{y}$. ADDED FOR CLARITY

Exam trap. For x^y: with respect to x it is a power, $y\,x^{y-1}$; with respect to y it is an exponential, $x^y \log x$. ADDED FOR CLARITY

Worked examples

W7.1 · First-order partials at a point

ESA ★★

GIVEN $f(x,y) = x^4 - x^2y^2 + y^4$ at $(-1, 1)$.

STEP 1 $f_x = 4x^3 - 2xy^2 = -4 + 2 = -2$.

STEP 2 $f_y = -2x^2y + 4y^3 = -2 + 4 = 2$.

STEP 3 Geometrically, the tangents parallel to the x- and y-axes at $(-1, 1, 1)$ have slopes -2 and 2.

ANSWER $\boxed{f_x(-1,1) = -2, \quad f_y(-1,1) = 2}$

CHECK Matches the answer printed on the CW sheet.

CW U1 P1(i)

W7.2 · First-order partials of a product

ESA ★★

GIVEN $f(x,y) = x^2 e^{-y/x}$ at $(4, 2)$.

STEP 1 $f_x = 2x\,e^{-y/x} + x^2 e^{-y/x}\cdot\dfrac{y}{x^2} = (2x + y)\,e^{-y/x} = 10\,e^{-1/2}$.

STEP 2 $f_y = x^2 e^{-y/x}\cdot\left(-\dfrac{1}{x}\right) = -x\,e^{-y/x} = -4\,e^{-1/2}$.

ANSWER $\boxed{f_x = 10e^{-1/2}, \quad f_y = -4e^{-1/2}}$

CHECK Matches the CW answer key.

CW U1 P1(ii)

W7.3 · Verify that the mixed partials are equal

ESA ★★

GIVEN $f(x,y) = x^{y}$, $x > 0$.

STEP 1 $f_x = y\,x^{y-1}$ (power rule); $f_y = x^{y}\log x$ (exponential rule).

STEP 2 $f_{xy} = \dfrac{\partial}{\partial y}\left(y\,x^{y-1}\right) = x^{y-1} + y\,x^{y-1}\log x = x^{y-1}(1 + y\log x)$.

STEP 3 $f_{yx} = \dfrac{\partial}{\partial x}\left(x^{y}\log x\right) = y\,x^{y-1}\log x + x^{y}\cdot\dfrac{1}{x} = x^{y-1}(1 + y\log x)$.

ANSWER $\boxed{f_{xy} = f_{yx} = x^{y-1}(1 + y\log x)}$

CHECK Both sides are identical, as Clairaut's theorem predicts.

CW U1 P2

W7.4 · Third-order partials at a point

ESA ★★

GIVEN $f(x,y) = e^{x}\log y + \cos y\log x$ at $(1, \pi/2)$. Find f_{xxx}, f_{xyy}, f_{yyy}.

STEP 1 $f_x = e^{x}\log y + \dfrac{\cos y}{x}$, $f_{xx} = e^{x}\log y - \dfrac{\cos y}{x^{2}}$, $f_{xxx} = e^{x}\log y + \dfrac{2\cos y}{x^{3}}$.

STEP 2 $f_{xy} = \dfrac{e^{x}}{y} - \dfrac{\sin y}{x}$, $f_{xyy} = -\dfrac{e^{x}}{y^{2}} - \dfrac{\cos y}{x}$.

STEP 3 $f_y = \dfrac{e^{x}}{y} - \sin y\log x$, $f_{yy} = -\dfrac{e^{x}}{y^{2}} - \cos y\log x$, $f_{yyy} = \dfrac{2e^{x}}{y^{3}} + \sin y\log x$.

STEP 4 At $(1, \pi/2)$: $\cos y = 0$, $\sin y = 1$, $\log 1 = 0$.

ANSWER $\boxed{f_{xxx} = e\log\dfrac{\pi}{2}, \quad f_{xyy} = -\dfrac{4e}{\pi^{2}}, \quad f_{yyy} = \dfrac{16e}{\pi^{3}}}$

CHECK Matches the CW key; it also gives $f_{xxy} = 1 + 2e/\pi$.

ESA Dec-23 1a · CW U1 P5

W7.5 · Value of n from a second-order equation

ESA ★★

GIVEN $\theta = t^{n}e^{-r^{2}/4t}$. Find n so that $\dfrac{1}{r^{2}}\dfrac{\partial}{\partial r}\left(r^{2}\dfrac{\partial\theta}{\partial r}\right) = \dfrac{\partial\theta}{\partial t}$.

STEP 1 $\dfrac{\partial\theta}{\partial r} = -\dfrac{r}{2}\,t^{n-1}e^{-r^{2}/4t}$, so $r^{2}\dfrac{\partial\theta}{\partial r} = -\dfrac{r^{3}}{2}\,t^{n-1}e^{-r^{2}/4t}$.

STEP 2 $\dfrac{1}{r^{2}}\dfrac{\partial}{\partial r}(\cdot) = \left(-\dfrac{3}{2}\,t^{n-1} + \dfrac{r^{2}}{4}\,t^{n-2}\right)e^{-r^{2}/4t}$.

STEP 3 $\dfrac{\partial\theta}{\partial t} = \left(n\,t^{n-1} + \dfrac{r^{2}}{4}\,t^{n-2}\right)e^{-r^{2}/4t}$.

STEP 4 Compare the t^{n-1} terms: $n = -\tfrac{3}{2}$.

ANSWER $\boxed{n = -\dfrac{3}{2}}$

CHECK The $r^{2}t^{n-2}$ terms agree on both sides for every n.

ESA Jan-24 s4 1c

Past-paper questions

ESA Dec 2023 · 6 marks · UE22MA141A Q1.a · solved in W7.4

If $f(x, y) = e^x \log y + \cos y \log x$, find f_{xxx}, f_{xyy} and f_{yyy} at $(1, \pi/2)$.

ESA Jan 2024 · 6 marks · UE23MA141A set 4, Q1.c · solved in W7.5

If $\theta = t^n e^{-r^2/4t}$, what value of n will make $\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial \theta}{\partial r}\right) = \frac{\partial \theta}{\partial t}$?

short-note recap

To find partials at a point, differentiate fully, holding the other variable constant, and substitute only at the end. To verify $f_{xy} = f_{yx}$, compute each side separately and compare. Watch the rule used: x^y is a power in x but an exponential in y.

A Symbol glossary

Symbol	Name	Meaning	Appears in
f, z	Function / dependent variable	$z = f(x, y)$; its graph is a surface	E1.1–E6.2
x, y	Independent variables	Coordinates in the xy-plane	all
(x_0, y_0)	Base point	Point where the derivatives are evaluated	E4.1–E5.2
P, z_0	Point on the surface	$P = (x_0, y_0, z_0)$ with $z_0 = f(x_0, y_0)$	E4.3
∂	“Partial”, read “dou”	Differentiate one variable, hold the others constant	E1.1–E1.4
f_x, f_y	First-order partials	Slopes parallel to the x- and y-axes	E1.1, E4.1–E5.2
f_{xx}, f_{yy}	Pure second-order partials	Concavity along x and along y	E1.2, E6.1
f_{xy}, f_{yx}	Mixed second-order partials	Change of one slope along the other direction	E1.2, E1.4, E6.2
m, n	Counts	m = number of variables, n = order	E1.3
$f'(x_0)$, dy/dx	Ordinary derivative	Slope of a tangent to a plane curve	E2.1, E2.2
∇f	Gradient (“del f”)	$f_x\,\mathbf{i} + f_y\,\mathbf{j}$	E5.1, E5.2
$\mathbf{i}$, $\mathbf{j}$	Unit vectors	Along the x- and y-axes	E5.1
$\hat{u} = (a, b)$	Unit direction vector	$a^2 + b^2 = 1$	E5.2
$D_{\hat{u}} f$	Directional derivative	Slope of the surface in direction $\hat{u}$	E5.2
t	Time (in F14)	Independent variable in the motion analogy	E6.3

B Sources, coverage and gaps

Source	Path	Use
Deck 002 (.pptx, rendered to images; .slides.md; .slides.txt)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/002_Geometrical_interpretation_of_Partial_derivatives_Problems_on_partial_derivatives__714d38d7a06bef55.*	11 slides; content on slides 3–10
Transcript 002 (.transcript.md / .json / .srt / .txt)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/002_Geometrical_interpretation_of_Partial_derivatives__374c7bb76e445af3.*	15.7 min lecture
Video 002 (.mp4): 3 single frames at 00:07:45, 00:09:05, 00:12:40	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/002_...374c7bb76e445af3.mp4	Remote seek; the full video was not downloaded
Unit 1 & 2 class-work sheet (UE24MA141A)	pesuc/First_Year_8enterprise/Files/Math 1/CW/Math_Unit1&2_CW__1jKDN6W8iw.pdf	Problems 1, 2, 5 used
ESA papers, UE22MA141A: Dec 2023, Jul 2023, May 2024	pesuc/First_Year_8enterprise/Files/Math 1/Papers/ESA Papers/UE22MA141A/*.pdf	Q1 read for Unit 1
ESA papers, UE23MA141A: Jan sets 1–4, June 2024	pesuc/First_Year_8enterprise/Files/Math 1/Papers/ESA Papers/UE23MA141A/*.pdf	Q1 read for Unit 1

Eight past ESA papers were checked: UE22MA141A Dec 2023, May 2024 and Jul 2023, and UE23MA141A Jan sets 1–4 and June 2024. None asks for the geometrical interpretation directly. Two 6-mark Q1 parts test higher-order partials, the computational half of this session. Other Unit 1 questions (Euler, Taylor, maxima, Lagrange, chain rule) belong to later sessions and are left out.

Known gaps, conflicts and corrections

1. The deck's title says “Problems on partial derivatives”, but the deck has no problem slides, and the transcript defers problems to the next class (T002 00:14:59). Problems were taken from the class-work sheet and ESA papers instead.
2. Slide 10 shows an interactive “Second-Order Partial Derivatives” demonstration. The transcript names its source unclearly (“Ultra Am lab”), and the video frame at 00:12:40 does not show a name, so the tool is described without one.
3. The link promised “in my next slide” (T002 00:15:16) is not in the deck; slide 11 is the closing slide.
4. Slides 4, 5 and 8 use third-party images (a YouTube source is credited on the slide). They were redrawn as Figures G2.1, G3.1, G3.2 and G5.1 rather than copied.
5. The UE23MA141A papers are headed “JAN 2023”, but the browser print header says Jan 2024. They are cited as Jan 2024.

6. Transcript recognition errors were corrected silently: “geometally” → geometrically, “Z1” → z_y, “impartial derivatives” → in partial derivatives, “Claret theorem” → Clairaut's theorem, “jowl” → jolt (another name for jerk).

Coverage

Source	Citations	Slides cited (slide × citations)
S002	75	3×13, 4×7, 5×5, 6×7, 7×12, 8×12, 9×17, 10×2
CW	9	
ESA	4	
T002	69	
V002	8	

Slides 1, 2 and 11 are the title, session and closing slides and hold no content.